

ESA KHAN

Abbottabad, KPK, Pakistan | +92-325-0070313 | esak62622@gmail.com | linkedin.com/in/esa-khan | github.com/esakhan786

Professional Summary

Machine Learning Engineer and Computer Science student (COMSATS University, 2027) specializing in LLMs, RAG Pipelines, Agentic AI, and Generative AI. Experienced in building end-to-end AI-powered applications using LangChain, LangGraph, Hugging Face, FastAPI, and Streamlit. Passionate about solving real-world problems with intelligent automation, conversational AI, and scalable NLP systems. Seeking internship or entry-level ML/AI Engineering roles at forward-thinking organizations.

Technical Skills

AI/ML Frameworks: LangChain, LangGraph, Groq API, OpenAI API, Hugging Face Transformers, TensorFlow, PyTorch, Scikit-learn

Specializations: Large Language Models (LLMs), Agentic AI, RAG Pipelines, NLP, Deep Learning, Generative AI

Backend & APIs: FastAPI, Flask, Streamlit

Languages: Python (Advanced), SQL

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Docker (basic), Linux

Databases: Vector Databases (FAISS, Chroma)

Projects

AI-Powered Self-Correcting Code Assistant | *Python, LangChain, Groq API, Streamlit* 2026

- Built an intelligent coding assistant that autonomously detects, analyzes, and corrects code errors using LLM-based reasoning loops and LangChain agentic workflows, reducing debugging time significantly.
- Implemented self-correction pipeline leveraging chain-of-thought prompting and iterative feedback loops to improve code quality across multiple languages.
- Exposed via Streamlit interface enabling seamless integration with developer workflows.

AI Video Agent for Audio Understanding & RAG-based Q&A | *LangChain, Whisper, Groq API, FAISS, Streamlit* 2026

- Developed a multimodal AI agent capable of processing video audio streams, transcribing speech using Whisper, and enabling natural language Q&A over video content via a RAG pipeline.
- Designed vector store indexing with FAISS to enable low-latency semantic retrieval over large audio transcripts.
- Built end-to-end Streamlit interface for real-time user interaction, audio chunking, and context-aware answer generation.

AI Attendance System via Voice & Face Recognition | *Deep Learning, OpenCV, FaceNet, SpeechRecognition, Python* 2026

- Built a dual-modal attendance system using deep learning face recognition (FaceNet/DeepFace) and voice verification for contactless, real-time automated attendance marking.
- Trained CNN-based face detection models with OpenCV and integrated voice recognition pipelines to authenticate users through biometric identity verification.
- Designed backend logging attendance upon successful face + voice match, achieving high accuracy in controlled environments.

FYP: AI Gig Generator & Resume-Based LinkedIn Job Matcher (In Progress) | *RAG, LLMs, KNN, Cosine Similarity, LangChain, LangGraph, FAISS, Streamlit* 2026

- Final Year Project currently in progress: AI platform combining gig content generation for Upwork/Fiverr and resume-based matching against live LinkedIn job listings, powered by RAG pipelines with KNN retrieval and cosine similarity re-ranking.
- Integrating LLMs for intelligent gig proposal generation and job description analysis, with FAISS vector indexing enabling scalable semantic search across live job postings scraped from LinkedIn in real time.
- LangGraph agentic loops iteratively refine gig content based on user skill level, niche, and feedback; project ongoing with core pipeline under active development.

Education

Bachelor of Computer Science (BCS) | COMSATS University Islamabad, Abbottabad Campus | Abbottabad, KPK 2023 – 2027

- Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Systems, OOP, AI, Computer Networks
- Honor: Campus Honor Roll – Academic Excellence

FSC – Pre-Engineering / Computer Science | Govt Postgraduate College Lakki Marwat | Lakki Marwat, KPK 2020 – 2022

Certifications & Achievements

- Huawei HCIA-AI V3.5 (x2) – Covers ML fundamentals, deep learning, and AI engineering | 2024
- Deep Learning Workshop Certificate – Awarded by COMSATS University Director for participation and excellence
- Campus Honor Roll – Academic Excellence Award, COMSATS University Abbottabad
- Open Source Contributor – Active GitHub contributor with AI/ML projects at github.com/esakhan786